IG for T

T	Yes	NO	Total	Entropy
Hot	1	1	2	1
mild	1	1	2	1
Cool	1	0	1	0

$H(S/Hot) = -\frac{1}{2}\log_2(1/2) - 1/2\log_2(1/2) = 1$

$H(S/mild) = 1$

$H(S/cool) = 0$ — IG for H —

H	Total	Yes	NO	Entropy
High	4	2	2	1
Normal	1	1	0	0

$H(S/high) = -\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{T}\right)$

Normal = 1

$H(S/wild) = 0.000.$

— IG for W —

W	Total	Yes	NO	Entropy
weak	4	3	1	0.811
Strong	1	0	1	0

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	OL	Temp	Humidity	Wind	Rain
1	Sunny	Hot	high	Weak	NO
2	Sunny	Wild	high	Strong	NO
3	Cloudy	Hot	high	weak	Yes
4	Rainy	Wild	high	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes

Entropy:

Yes = 3

$H(S) = -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)$

$H(S) = +0.442 + 0.528$

$H(S) = 0.97$

OL	Yes	NO	Total	Entropy
Sunny	0	2	2	0.00
Cloudy	1	0	1	0.00
Rainy	2	0	2	0.00

$H(S/sunny) = -\frac{0}{2}\log_2\left(\frac{0}{2}\right) - \frac{2}{2}\log_2($

$H(S/cloudy) = 0 - 0$

$H(S/Rainy) = 0$

$H(S/\text{weak}) = -\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}$

$\log\left(\frac{3}{4}\right) = 0.811$

$H(S/\text{strong}) = 0.00$

HS Calculation

$H(S/OC) = \frac{2}{5}(0) + \frac{1}{5}(0) + \frac{2}{5}(0)$

$= 0.8$

$H(S/H) = \frac{4}{8}(1) + \frac{1}{5}(0)$

$= 0.8$

$H(S/w) = \frac{4}{5}(0.84) + \frac{1}{5}(0)$

$= 0.811$

IG :-

$IG(OL) = 0.97 - 0 = 0.97$

$IG(OT) = 0.97 - 0.8 = 0.17$

$IG(H) = 0.97 - 0.8 = 0.17$

$IG(w) = 0.97 - 0.811 = 0.15$

OL is root node.

